MINI PROJECT REPORT

AIR QUALITY ANALYSIS OF 26 INDIAN CITIES

A data analysis study of daily Air Quality Index readings
(1 January 2015 – 1 July 2020)

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Abstract

In this project, we looked at air quality in 26 Indian cities using daily Air Quality Index (AQI) data from 1 January 2015 to 1 July 2020. The dataset contains 29,531 city-day records and measurements for twelve pollutants, along with AQI. We used Python to check and clean the data, create date-related columns, compare cities, look at seasonal changes, and study how pollutant levels are related to AQI. We also compared AQI during the same period in 2019 and 2020 to see what changed around the 2020 lockdown. The results are shown in an interactive HTML dashboard. In the dashboard summary, Ahmedabad has the highest average AQI and Aizawl has the lowest. Winter has the highest seasonal average, while the overall fitted trend moves downward over the years.

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1. Introduction

Air pollution is a concern because the air quality people experience can change from one city to another and from season to season. The Air Quality Index (AQI) makes pollution levels easier to understand by expressing them as a number and placing them into categories. In this project, we use historical daily readings from Indian cities to compare their AQI, look for changes during different months and seasons, and check how AQI is related to the measured pollutants.

We carried out the data cleaning and analysis in a Python notebook and used an HTML dashboard to make the results easier to explore.

2. Problem Statement

The main aim of this project is to study past air-quality readings from different Indian cities. We focus on how pollution levels differ between cities, how they change with the seasons, how AQI is related to pollutant measurements, and what differences appear in the selected 2020 lockdown-period comparison.

3. Objectives

- Load the city_day.csv dataset and understand its structure.
- Check for missing values in the pollutant and AQI columns and handle them.
- Compare the average AQI of the cities in the dataset.
- Look at how AQI changes across years and months.
- Group daily readings into the relevant AQI categories.
- Compare air quality across the four seasons.
- Study the relationship between AQI and the pollutant measurements.
- Compare AQI for 25 March–30 June in 2019 and 2020.
- Calculate a straight-line trend for the national daily average AQI.
- Show the analysis through an interactive dashboard.

4. Dataset Description

For this analysis, we used the file city_day.csv. It contains 29,531 daily records from 26 Indian cities, covering the period from 1 January 2015 to 1 July 2020, as described in the dashboard.

Dataset field / group	Description
City	Name of the Indian city.
Date	Date of the daily observation.
AQI	Air Quality Index value.
Pollutants	PM2.5, PM10, NO, NO2, NOx, NH3, CO, SO2, O3, Benzene, Toluene and Xylene.
AQI_Bucket	Category derived from AQI values.
Derived fields	Year, Month and Season created from Date.

The amount of data available is not the same for every city. Some cities have readings for only part of the full period. Because of this, city comparisons should be considered along with the amount of data available for each city.

5. Tools and Technologies

Tool / library	Purpose in the project
Python	Main programming language for data analysis.
Pandas	Loading CSV data, cleaning, grouping, aggregation and table operations.
NumPy	Numerical operations and linear trend fitting using polyfit.
Matplotlib	Plotting charts.
Seaborn	Statistical visualizations such as heatmaps, boxplots and scatterplots.

Requests	Sending an HTTP request for the population-data scraping step.
BeautifulSoup	Imported in the notebook for web-related processing.
Google Colab	Notebook environment and file upload/download workflow.
HTML, CSS and JavaScript	Interactive dashboard interface and city comparisons.

6. Methodology

6.1 Data loading and inspection

We first imported the required Python libraries and uploaded city_day.csv in Google Colab. The Date column was converted into a date format. We then checked the dataset size, date range, number of cities, missing values and basic statistics to understand the data before starting the analysis.

6.2 Data cleaning

1. First, we sorted the records by city and date.
2. For the pollutant and AQI columns, we filled missing values by interpolating within each city. This step was limited to gaps of up to seven consecutive values, and interpolation was applied in both directions.
3. If any values were still missing, we filled them with the median value for that city.
4. After these steps, we removed any rows where AQI was still missing.
5. Finally, we recreated the AQI category for each row using its AQI value.

The notebook uses these AQI ranges: Good (up to 50), Satisfactory (above 50 to 100), Moderate (above 100 to 200), Poor (above 200 to 300), Very Poor (above 300 to 400), and Severe (above 400).

6.3 Feature engineering

We used the Date column to create separate Year and Month columns. We also added a Season column based on the month, as shown below:

Season	Months
Winter	December, January, February
Summer	March, April, May
Monsoon	June, July, August, September
Post-monsoon	October, November

6.4 Exploratory analysis

To explore the data, we grouped records by city, year, month, season and date, and calculated the average AQI for each group. We also checked the share of records in each AQI category, calculated pollutant correlations, and worked out the percentage change for the lockdown-period comparison.

6.5 National trend

For the national trend, we first calculated the average AQI across cities for each date. We then converted the elapsed time into years and used NumPy's polyfit function to fit a straight line to the daily national average.

7. Analysis and Results

7.1 City-wise AQI comparison

The city comparison in the dashboard shows that Ahmedabad has the highest average AQI at 426.7, while Aizawl has the lowest at 34.8. This shows how much the average AQI can differ between the cities included in the dataset.

Measure	Result shown in dashboard
Highest mean AQI	Ahmedabad – 426.7
Lowest mean AQI	Aizawl – 34.8
Delhi mean AQI in city profile	259.5
Patna mean AQI in city profile	240.8
Lucknow mean AQI in city profile	218.0

7.2 Yearly and monthly patterns

We calculated the yearly average AQI for the five cities with the highest overall averages and displayed the results in a line chart. We also created a heatmap of average AQI by city and month. These charts make it easier to spot months and years when AQI was relatively higher or lower. The dashboard includes these views so they can be explored further.

7.3 AQI category distribution

We divided the daily readings into six AQI categories and calculated the percentage of city-day records in each category. The results are displayed in a bar chart, giving an idea of how often each air-quality category appears in the dataset.

7.4 Seasonal analysis

According to the dashboard, Winter has the highest seasonal average AQI at 214.4, while Monsoon has the lowest at 121.3. We also used a boxplot to compare the spread of AQI values across seasons. Outlier markers were hidden to keep the chart easier to read.

Seasonal result	Mean AQI
Winter – highest	214.4
Monsoon – lowest	121.3

7.5 Pollutant relationship

We calculated the correlation between AQI and the twelve pollutant measurements. In the dashboard results, Carbon Monoxide (CO) and PM2.5 have the strongest positive correlations with AQI, at 0.63 and 0.62 respectively.

This means they show a relatively strong relationship with AQI in this dataset, but correlation alone does not show that one factor causes the other.

Pollutant	Correlation with AQI
CO	0.63
PM2.5	0.62

We also created a scatterplot using a sample of 3,000 records to see how PM2.5 and AQI are related. The points are grouped by season.

7.6 Lockdown comparison

For the lockdown comparison, we compared each city's average AQI from 25 March to 30 June in 2019 with the average for the same dates in 2020. We calculated the percentage change for cities that had data in both periods. The dashboard shows a lower AQI in 19 out of the 20 cities compared. Ahmedabad had the largest reported decrease, at 76.3%.

This comparison looks at the same date range in two different years. It gives us a way to examine the change, but it does not prove that the lockdown was the only reason for it.

7.7 National AQI trend

The dashboard's fitted national trend shows a decrease of 13.3 AQI points per year over the period studied. It also points out that some of this decline appears during the lockdown months near the end of the dataset.

8. Interactive Dashboard

The project also includes a browser-based dashboard in the file index.html. It is titled “How polluted is the air in India’s cities?” and brings the main results together in one place. Its sections include:

- An overview with the main indicators and summary results.
- City profiles showing average AQI, PM2.5, category percentages and monthly AQI.
- A comparison tool for viewing two cities side by side.
- A year explorer and a section about data quality.
- City rankings, yearly trends and a monthly heatmap.
- Charts for AQI categories, seasons and pollutant relationships.
- The lockdown-period comparison and national AQI trend.
- A conclusion section with the main findings.

When a user selects a city, the city profile and city comparison sections update their values using JavaScript. The dashboard also includes a cigarette-equivalent display, but clearly describes it as a rough illustration based on a rule of thumb, not as a medical measurement.

9. Key Findings

- Ahmedabad has the highest average AQI in the dashboard summary (426.7), and Aizawl has the lowest (34.8).
- Winter has the highest seasonal average AQI (214.4), while Monsoon has the lowest (121.3).
- CO (0.63) and PM2.5 (0.62) show the strongest positive correlations with AQI among the reported pollutants.

- In the selected lockdown-period comparison, AQI is lower in 19 of the 20 cities in 2020 than in the matching period in 2019.
- Ahmedabad has the largest reported reduction in that comparison, at 76.3%.
- The fitted national AQI trend decreases by 13.3 points per year.

10. Limitations

- The analysis only covers the cities, dates and readings available in city_day.csv. The amount of data available differs between cities.
- Interpolation and filling missing values with city-wise medians help complete the dataset, but these choices may affect averages and correlations.
- The lockdown comparison uses matching date ranges from two different years. It cannot separate the effect of lockdown from other factors that may have changed.
- A correlation shows that two measurements are related; it does not prove that one causes the other.
- The national trend is based on a straight-line fit over a limited historical period, so it should not be used as a forecast.
- The notebook includes a step for collecting city population data from Wikipedia. However, the conclusions in the supplied dashboard do not include an AQI analysis adjusted for population.

11. Conclusion

Through this project, we used Python to study air quality across 26 Indian cities. We started by loading and cleaning the daily readings, then compared average AQI across cities, looked at monthly and seasonal patterns, and checked the relationship between AQI and different pollutants. We also compared a selected period in 2019 and 2020. The results show clear differences between cities and seasons. Ahmedabad has the highest reported average AQI, while Aizawl has the lowest. Winter has the highest seasonal average, and CO and PM2.5 have the strongest positive correlations with AQI in the dashboard. The interactive dashboard makes it easier to explore these results through city profiles, comparisons and charts.

12. Future Scope

- Include more recent data and, if possible, readings from additional Indian cities.
- Add weather information, such as temperature, rainfall and wind, to study how it relates to AQI.
- Try different ways of filling missing values and check how they affect the results.
- Develop an AQI forecasting model and test it on a separate time period before using it for predictions.
- Include population and location details after checking that the external data matches the city names in the AQI dataset.
- Publish the dashboard online and provide options to download city-wise summary tables.

13. References

- Project dataset: city_day.csv (as loaded by Mini_Project_SL.ipynb).
- Project implementation: Mini_Project_SL.ipynb (uploaded notebook).
- Project dashboard: index.html (uploaded dashboard file).

- Python library documentation: pandas, NumPy, Matplotlib and Seaborn.

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